

MENGLA, China

WMO: 569690

Lat: 21.4746N

Lon: 101.5780E

Elev: 633

StdP: 93.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 9.0	(c) 10.3	(d) 5.8	(e) 6.2	(f) 20.6	(g) 7.6	(h) 7.0	(i) 18.5	(j) 5.0	(k) 25.1	(l) 4.3	(m) 25.0	(n) 0.7	(o) 160	(p) 0.236

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 7.5	(c) 34.3	(d) 21.9	(e) 33.2	(f) 22.2	(g) 32.2	(h) 22.4	(i) 25.2	(j) 29.3	(k) 24.9	(l) 29.1	(m) 24.6	(n) 28.6	(o) 2.4	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.3	20.8	26.4	23.9	20.3	26.3	23.6	19.9	26.0	81.8	29.4	80.2	29.3	78.8	28.8	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
4.3	3.5	3.0	DB	7.2	36.2	1.7	1.2	5.9	37.1	4.9	37.8	3.9	38.5	2.7	39.3	
			WB	6.9	25.9	1.7	0.6	5.6	26.3	4.6	26.6	3.6	26.9	2.4	27.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	18.8	20.1	22.5	25.2	26.2	26.6	26.1	26.1	25.8	24.3	21.5	18.8
	DBStd	3.42	1.96	2.37	2.16	1.90	1.82	1.16	1.27	1.22	1.34	1.68	2.12	2.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	57	18	11	3	0	0	0	0	0	0	0	3	23
	CDD10.0	4934	273	282	389	457	503	499	498	499	473	443	344	274
	CDD18.3	1950	32	60	133	207	245	249	239	241	223	184	97	39
	CDH23.3	13662	213	570	1087	1675	2006	1961	1606	1548	1405	996	443	153
(14) Wind	WSAvg	1.1	1.0	1.1	1.1	1.1	1.2	1.3	1.1	1.1	1.1	1.0	0.9	0.9
	PrecAvg	1522	25	23	33	95	174	220	296	318	156	97	57	26
(16) Precipitation	PrecMax	1944	111	97	131	202	377	344	546	508	322	228	194	122
	PrecMin	1206	0	0	0	17	50	125	91	174	40	13	5	0
	PrecStd	195	27	24	28	48	70	68	101	90	74	52	53	29
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.5	31.9	33.8	35.9	36.5	34.2	33.2	33.1	32.7	31.7	30.2	28.2
		MCWB	17.3	17.6	19.0	20.5	22.1	23.7	24.3	24.3	23.7	22.9	21.3	20.2
	2%	DB	27.2	30.6	32.6	34.5	34.4	32.6	31.6	31.6	31.6	30.6	28.9	26.5
		MCWB	17.6	17.6	18.9	21.3	22.6	24.0	24.3	24.3	23.6	22.6	20.9	19.2
	5%	DB	25.9	29.2	31.4	33.1	32.8	31.4	30.5	30.4	30.6	29.6	27.6	25.2
		MCWB	17.7	17.7	19.1	21.7	23.0	24.0	24.3	24.2	23.5	22.3	20.5	18.7
	10%	DB	24.3	27.5	29.7	31.6	31.3	30.2	29.2	29.3	29.4	28.3	26.2	23.6
		MCWB	17.5	17.6	19.3	21.8	23.2	24.0	24.2	24.1	23.4	22.1	19.8	17.9
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	20.9	22.7	24.3	25.2	25.7	25.7	25.7	25.2	24.4	23.1	21.7
		MCDB	24.8	27.7	28.5	30.5	30.3	29.6	29.5	29.6	29.1	28.8	27.4	25.7
	2%	WB	19.3	19.9	21.7	23.6	24.6	25.1	25.2	25.1	24.7	23.7	22.2	20.5
		MCDB	24.0	26.0	27.9	29.9	29.7	29.4	29.1	29.1	28.8	28.0	26.2	24.3
	5%	WB	18.5	19.1	21.0	23.0	24.2	24.7	24.7	24.7	24.3	23.3	21.6	19.6
		MCDB	23.3	25.5	27.2	29.1	29.2	28.8	28.5	28.4	28.2	27.3	25.4	23.3
	10%	WB	17.8	18.2	20.2	22.5	23.7	24.4	24.4	24.4	24.0	22.9	20.8	18.8
		MCDB	22.5	24.4	26.3	28.4	28.6	28.3	27.9	27.8	27.7	26.5	24.3	22.2
(36) Mean Daily Temperature Range	5% DB	MDBR	12.2	15.2	14.7	12.8	9.9	7.5	6.5	6.8	7.9	8.6	10.1	10.4
		MCDBR	14.9	18.4	17.7	15.5	13.0	9.7	8.6	8.8	9.6	10.3	11.4	11.8
		MCWBR	5.9	6.2	5.3	4.0	3.2	2.4	2.5	2.6	2.7	3.2	4.1	4.7
	5% WB	MCDBR	12.3	15.0	14.0	12.8	9.9	7.7	7.4	7.5	8.1	8.6	9.5	10.2
		MCWBR	5.3	5.9	5.1	4.2	3.0	2.4	2.6	2.6	2.7	3.0	3.7	4.3
(40) Clear-Sky Solar Irradiance	taub		0.428	0.513	0.784	0.768	0.494	0.443	0.440	0.476	0.492	0.518	0.436	0.430
	taud		2.177	1.932	1.405	1.485	2.171	2.354	2.347	2.224	2.150	2.032	2.216	2.184
	Ebn at Noon		833	780	601	623	814	849	853	826	809	766	819	814
	Edh at Noon		136	185	326	304	152	126	127	143	152	165	130	131
	RadAvg		3.89	4.95	5.23	5.77	5.24	4.50	4.09	4.37	4.47	4.11	4.04	3.54
(45) All-Sky Solar Radiation	RadStd		0.31	0.52	0.38	0.51	0.53	0.37	0.39	0.36	0.22	0.25	0.31	0.31

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.31	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	-51	+175	+130

Nomenclature: See separate page